AI-Powered Hybrid Resume Analysis and Job-Matching System

PROJECT BATCH: 2

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Academic Batch : 2022 – 2026

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ABSTRACT

The increasing adoption of Applicant Tracking Systems (ATS) has automated resume screening but often fails to capture contextual skill relationships, disadvantaging fresh graduates and creating inefficiencies for recruiters managing large applicant pools. This work proposes an AI-powered hybrid resume analysis and job-matching system that integrates transformer-based sentence embeddings (SBERT), TF-IDF keyword similarity, and structured skill-gap analysis to improve contextual understanding. On the job seeker side, a weighted scoring mechanism combining semantic similarity, keyword matching, and skill coverage generates personalized job recommendations while identifying missing competencies and detecting experience level. Experimental evaluation shows improved contextual recommendations, with relevant junior-level roles achieving match scores above 35% and effective identification of critical skill gaps.

For recruiters, candidate profiles are encoded using Sentence-BERT embeddings and benchmarked across multiple machine learning models, including Logistic Regression, Linear SVM, XGBoost, and Random Forest. The Random Forest classifier achieved the highest performance with 92.41% accuracy in role classification. A dynamic semantic shortlisting engine ranks candidates using cosine similarity, achieving match scores above 61% for highly relevant data science and machine learning roles. The proposed system enhances transparency, reduces manual screening effort, and enables more accurate, data-driven recruitment decisions for both applicants and hiring teams.

OBJECTIVE

- Proposes an AI-powered hybrid resume analysis and job=matching system that combines NER-based information extraction with both statistical (TF-IDF) and semantic (BERT/SBERT) similarity techniques to overcome limitations of traditional ATS systems.
- To provide explainable and actionable feedback to job applicants by identifying skill gaps, weak areas, and resume improvement suggestions aligned with job requirements.
- To support recruiters with automated candidate ranking and analytics, reducing manual screening effort while improving fairness, transparency, and hiring efficiency.

BASE PAPER DETAILS

Title :AI-POWERED JOB APPLICATION MANAGEMENT FOR APPLICANTS

Published year :31st July 2025

Published in : Journal of Theoretical and Applied Information Technology

Authors :ZI QING CHEW1, NOR FAZLIDA MOHD SANI2

Link :https://zenodo.org/records/18106765

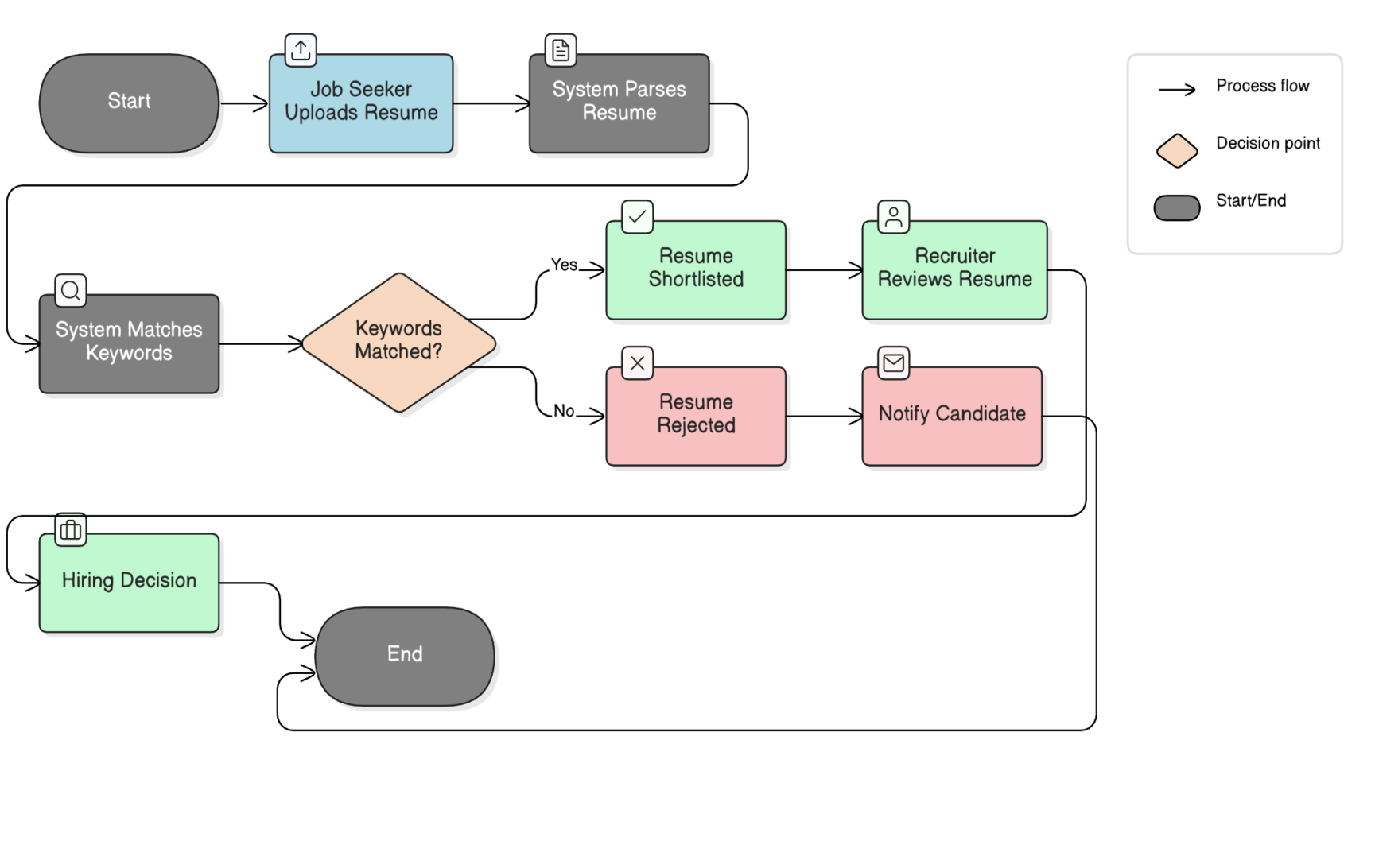
LITERATURE SURVEY(TABLE)

Title & Authors	Journal / Year	Dataset Used	Approach	Outcome	Limitations
NLP-Powered Resume Matching for Recruitment Isha Rathi et al.	IJFMR, 2024	PDF resumes & job descriptions (custom dataset)	TF-IDF, Cosine Similarity, NLP-based skill extraction, ML-based job-fit classification	Achieved ~85% accuracy in predicting job fit and effective resume-job matching	Uses predefined skill lists; TF-IDF fails to capture deep semantic context; limited handling of soft skills
Resume Parser with Natural Language Processing Sroison & Chan	Authorea Preprints, 2023	Large-scale online recruitment resumes	NLP-based resume parsing and job requirement matching	Reduced recruiter workload and improved consistency in candidate screening	Limited handling of ambiguous language; depends heavily on keyword relevance

LITERATURE SURVEY(TABLE)

Resume Parser Using NLP Kashif & Parimal Kumar K R	IJARCCE, 2024	Applicant resumes (varied formats)	NLP + ML with real-time resume feedback system	Improved resume quality and recruiter insights through automated analysis	Lacks deep contextual understanding; feedback quality depends on training data diversity
Resume Parser Analysis Using ML and NLP Rasal et al.	Conference Publication	Unstructured resumes	SVM, Random Forest, NLP preprocessing	Improved screening accuracy and scalability	Requires heavy preprocessing; performance drops with non-standard resume formats
A Machine Learning Approach for Automation of Resume Recommendation System Pradeep Kumar Roy et al.	Procedia Computer Science (Elsevier), 2020	Kaggle Resume Dataset (categorized resumes)	TF-IDF, Linear SVM, Cosine Similarity, k-NN	Linear SVM achieved ~78.5% accuracy; effective resume classification & ranking	Does not use deep learning; limited semantic understanding; industry-specific tuning required

EXISTING SYSTEM



Examples:

- Applicant Tracking Systems(ATS) (Workday, Taleo)
- Job portals with Basic Resume Matching (Naukri, Indeed, LinkedIn)
- Manual resume screening by recruiters

LIMITATIONS

EXISTING SYSTEM

- Over-reliance on exact keyword matching, causing qualified candidates to be rejected due to vocabulary or phrasing differences.
- Lack of semantic understanding, as traditional systems fail to capture contextual relationships between skills and job requirements.
- Poor support for fresh graduates, whose resumes may lack optimized formatting or explicit keywords despite relevant skills.
- Limited transparency and explainability, offering no clear justification for match scores or rejection decisions.
- More High manual effort and potential bias in recruiter-side screening when automated systems are inaccurate or inconsistent.

PROPOSED SYSTEM

- Develops an AI-driven resume intelligence system using transformer-based NER to convert unstructured resumes and job descriptions into structured, machine-readable profiles.
- Employs a hybrid relevance scoring framework that fuses TF-IDF statistical matching with BERT/SBERT semantic embeddings for context-aware and robust job–resume alignment.
- Integrates explainable AI mechanisms to generate interpretable match scores, skill-gap analysis, and targeted resume optimization recommendations.
- Implements a scalable, deployment-ready full-stack architecture enabling real-time parsing, batch candidate ranking, and data-driven recruitment analytics.

REQUIREMENTS

Software Requirements	Used Version
Google Colab	3.6
Visual Studio Code	1.72.2
Tensorflow	2.6.0

Hardware Requirements	Used Version
Processor	Intel Core i7-8550U CPU
RAM	8.00GB
HardDisk	1 TB

DATA COLLECTION

1. Candidate_Template_Final.csv

Type: Candidate Profile Dataset
Number of Records: 4,350
Features: First name, last name, current job title, key skills, notes, email
Usage: Candidate profiling, semantic embedding generation, role classification, and recruiter-side dynamic shortlisting

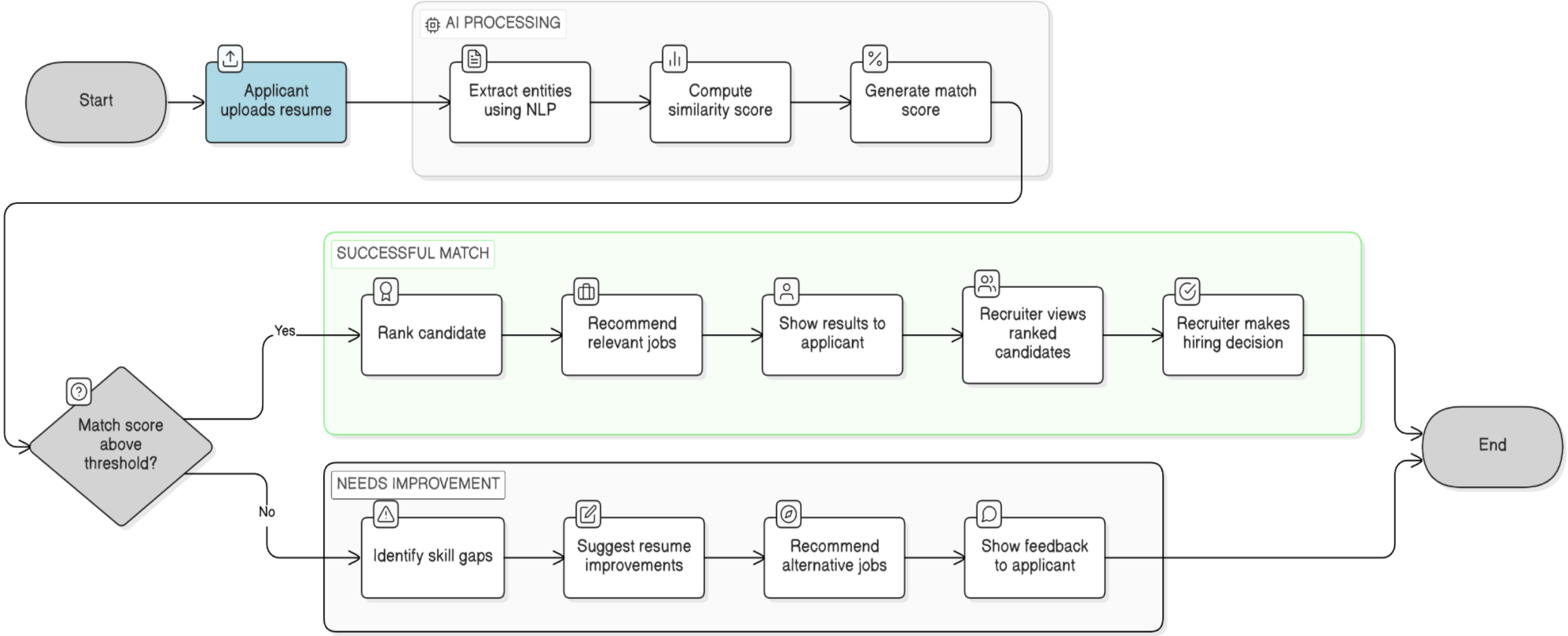
2. Job_Descriptions_Final_India.csv

Type: Job Descriptions Dataset
Number of Records: 5,800
Features: Job title, required skills, job description
Usage: Job requirement extraction, hybrid semantic matching (TF-IDF + SBERT), and job recommendation generation

3. IT_Job_Roles_Skills.csv(Kaggle)

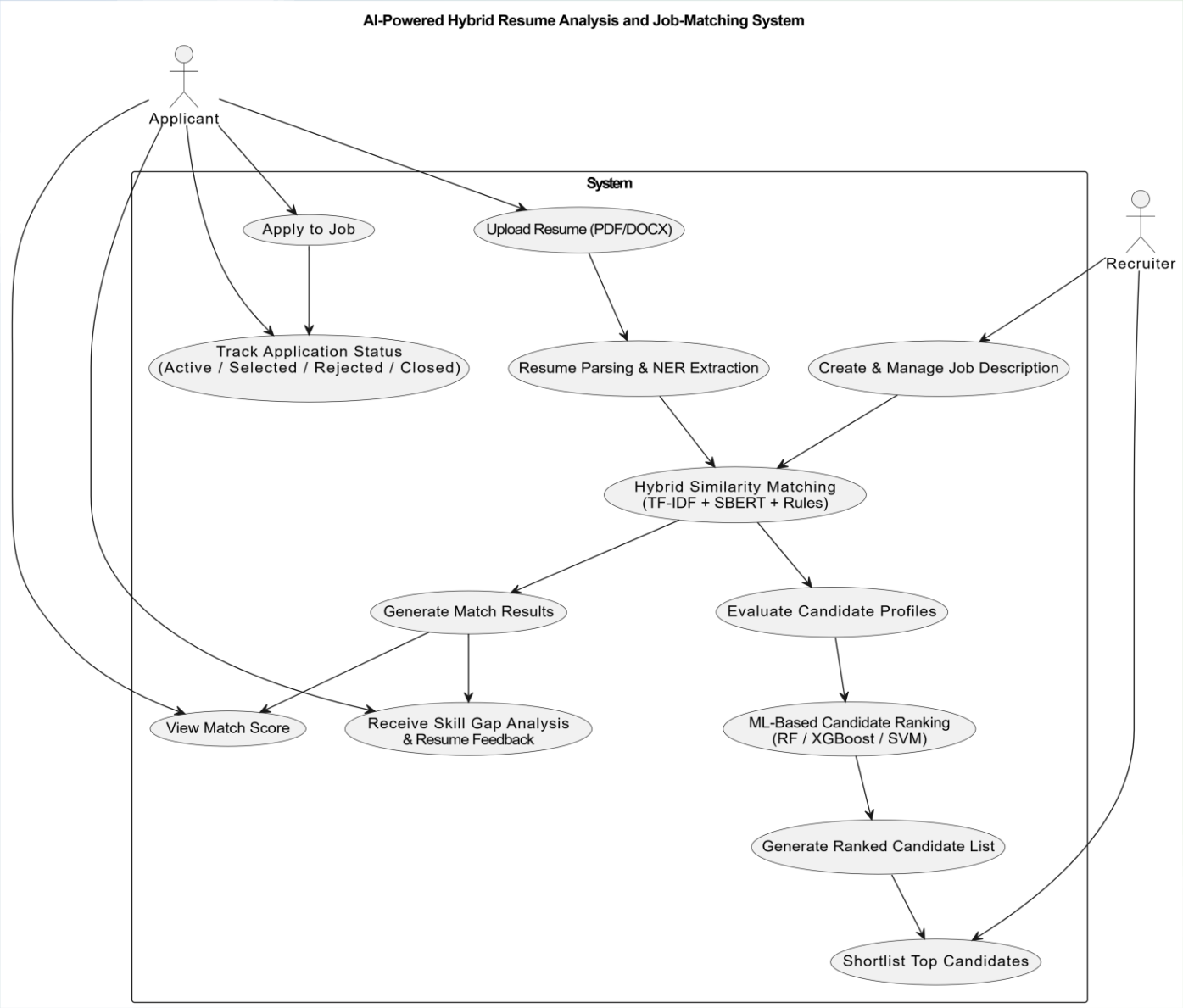
Type: Job Roles–Skills Mapping Dataset
Number of Records: 493
Features: Job role, associated technical skills
Usage: Skill extraction, role–skill alignment, and skill-gap analysis

SYSTEM OVERVIEW



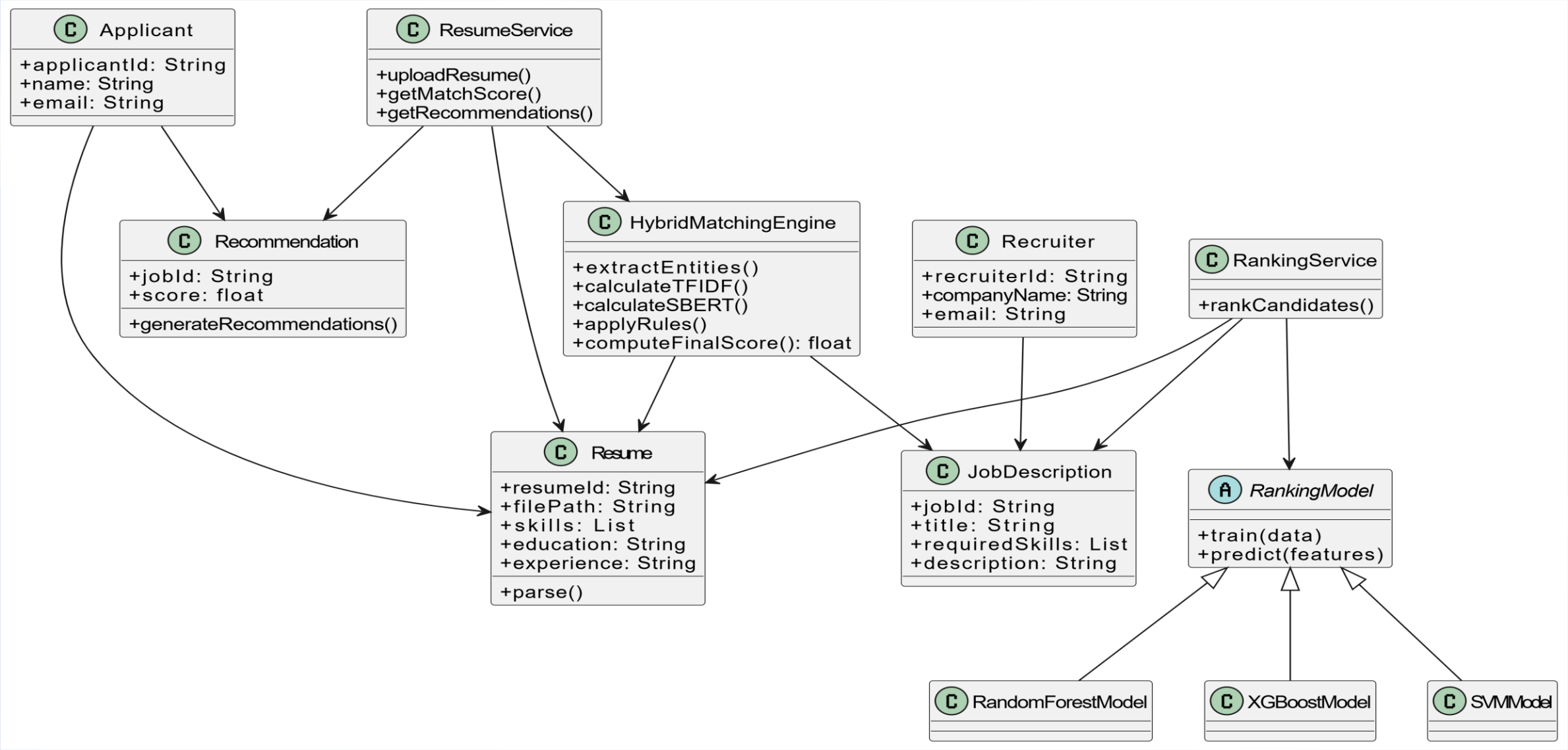
UML Diagrams

USE-CASE DIAGRAM



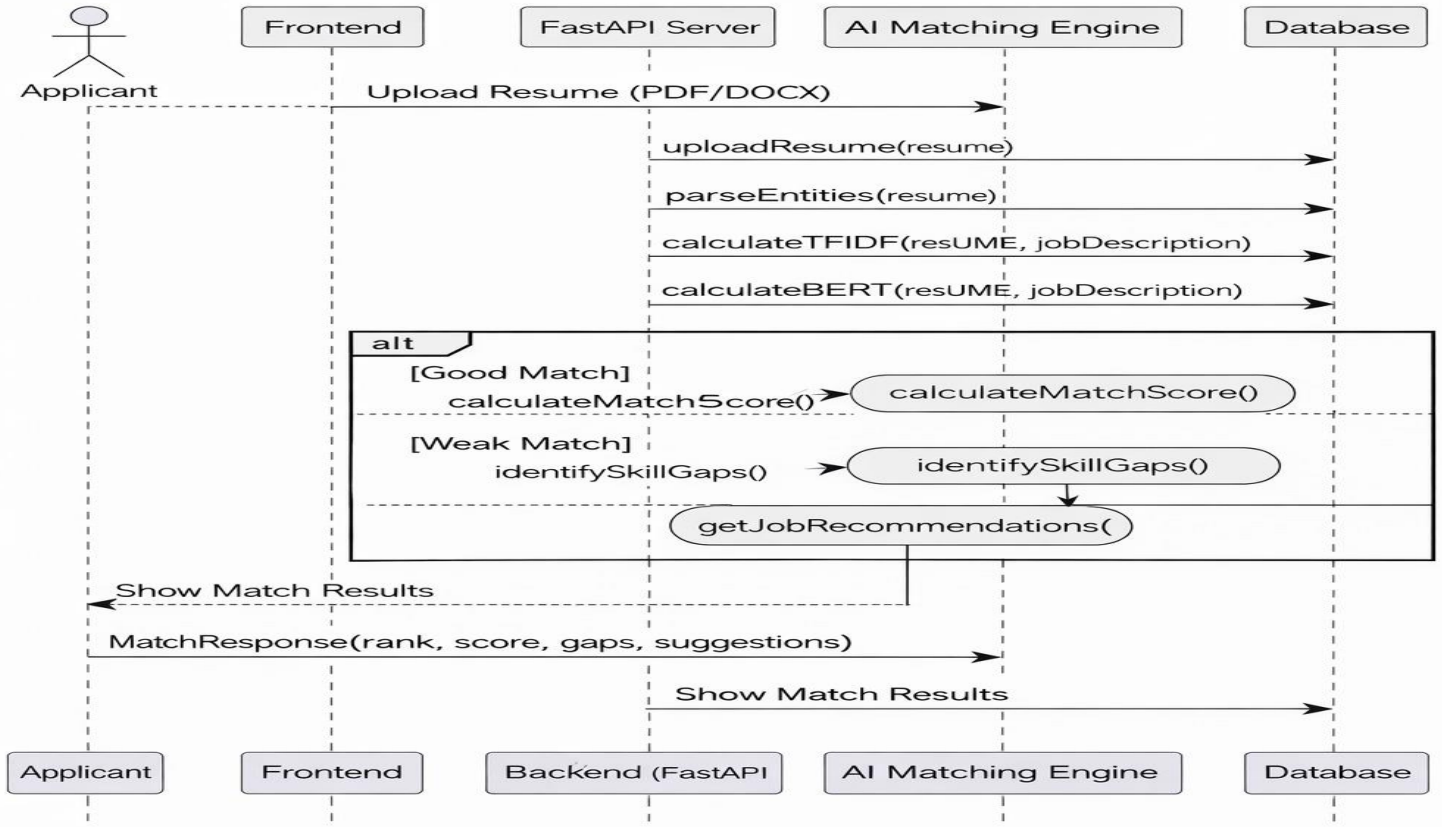
UML Diagrams

CLASS DIAGRAM



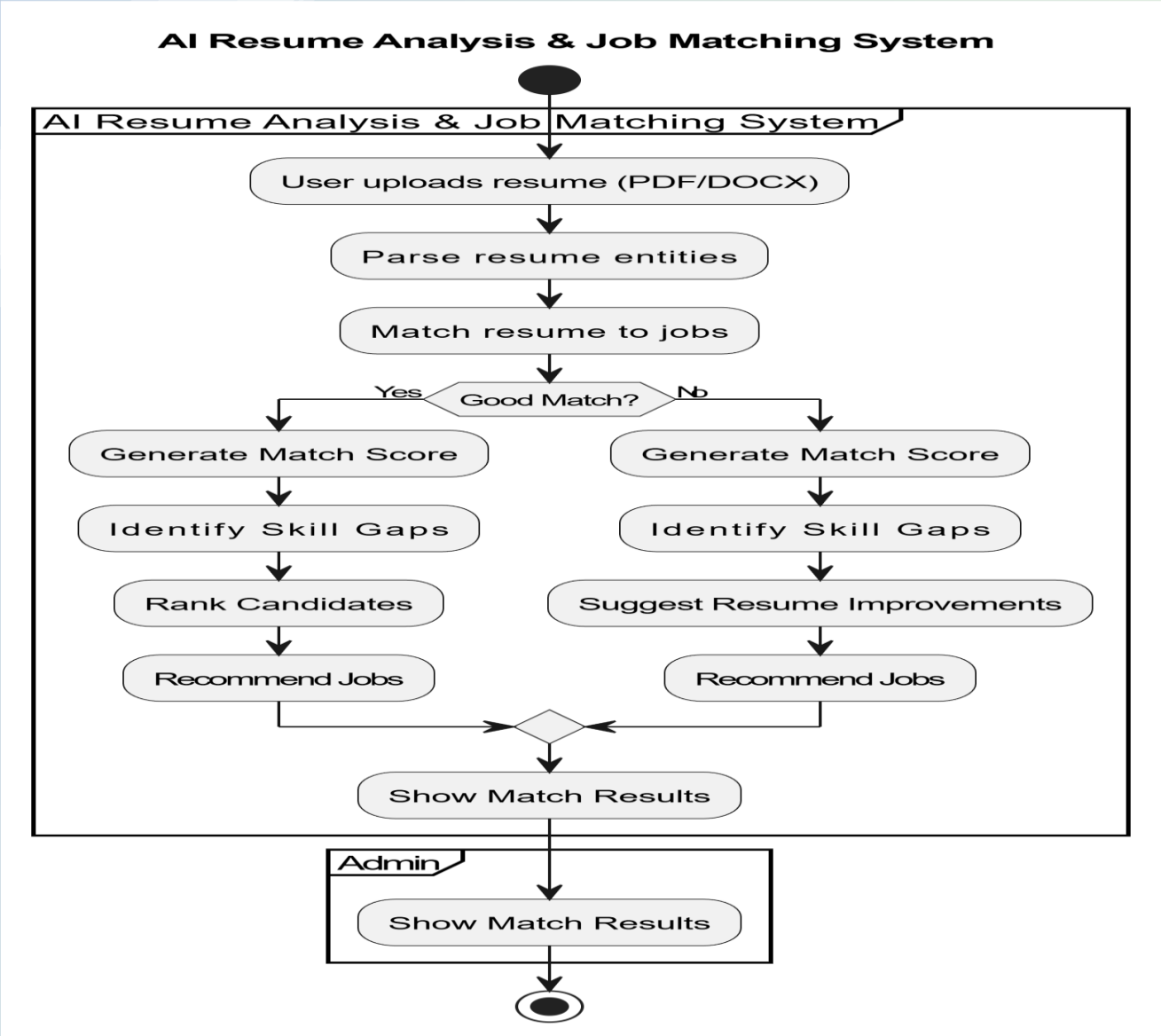
UML Diagrams

SEQUENCE DIAGRAM



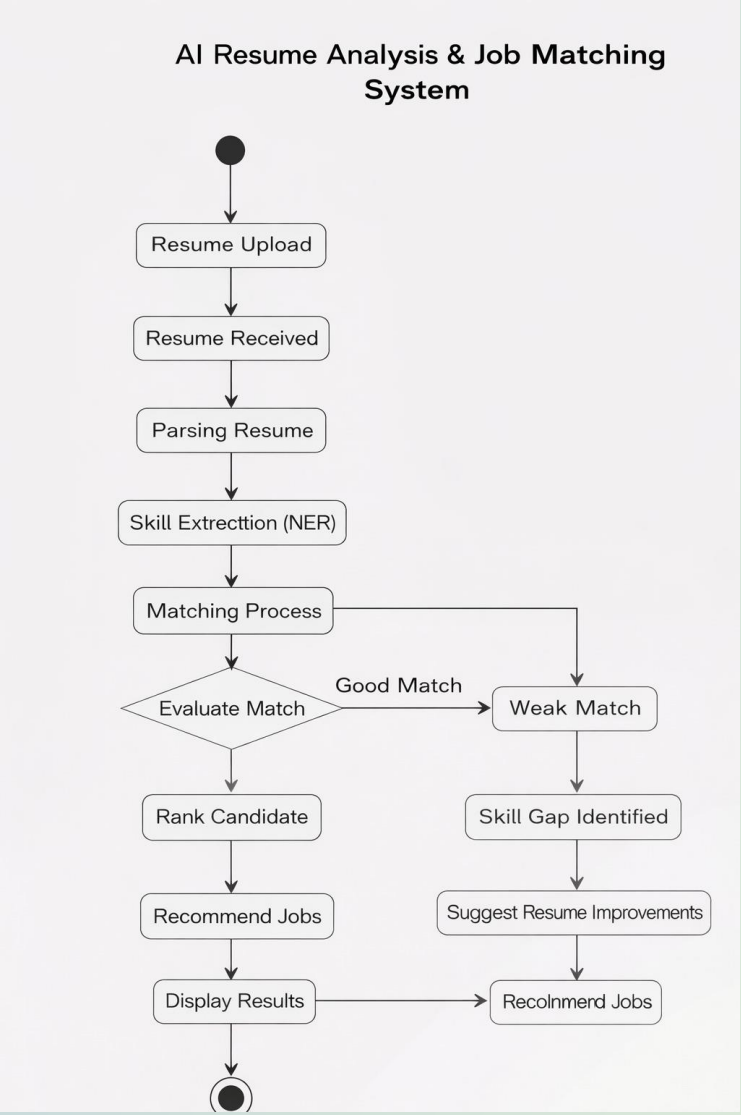
UML Diagrams

ACTIVITY DIAGRAM



UML Diagrams

STATE CHART DIAGRAM



RESULTS - RECRUITER SIDE MODEL

--- Model Accuracy ---
Logistic Regression: 44.83%
Random Forest: 92.41%
Linear SVM: 90.23%
XGBoost: 84.71%

🔎 Matching candidates for role:
👉 Data Scientist with Python, Machine Learning and NLP
==========

Final Shortlisted Candidates:

	Rank	First Name	Last Name	Current Job Title	Email \	Match Score (%)
0	1	Sneha	Nair	Junior Data Scientist	sneha.nair@gmail.com	61.23
1	2	Rohit	Gupta	Mid-Level AI Developer	rohit.gupta@gmail.com	59.47
2	3	Ankit	Gupta	Junior Data Scientist	ankit.gupta@gmail.com	59.11
3	4	Amit	Nair	ML Engineer (Fresher)	amit.nair@gmail.com	57.18
4	5	Rahul	Iyer	ML Engineer (Fresher)	rahul.iyer@gmail.com	55.77

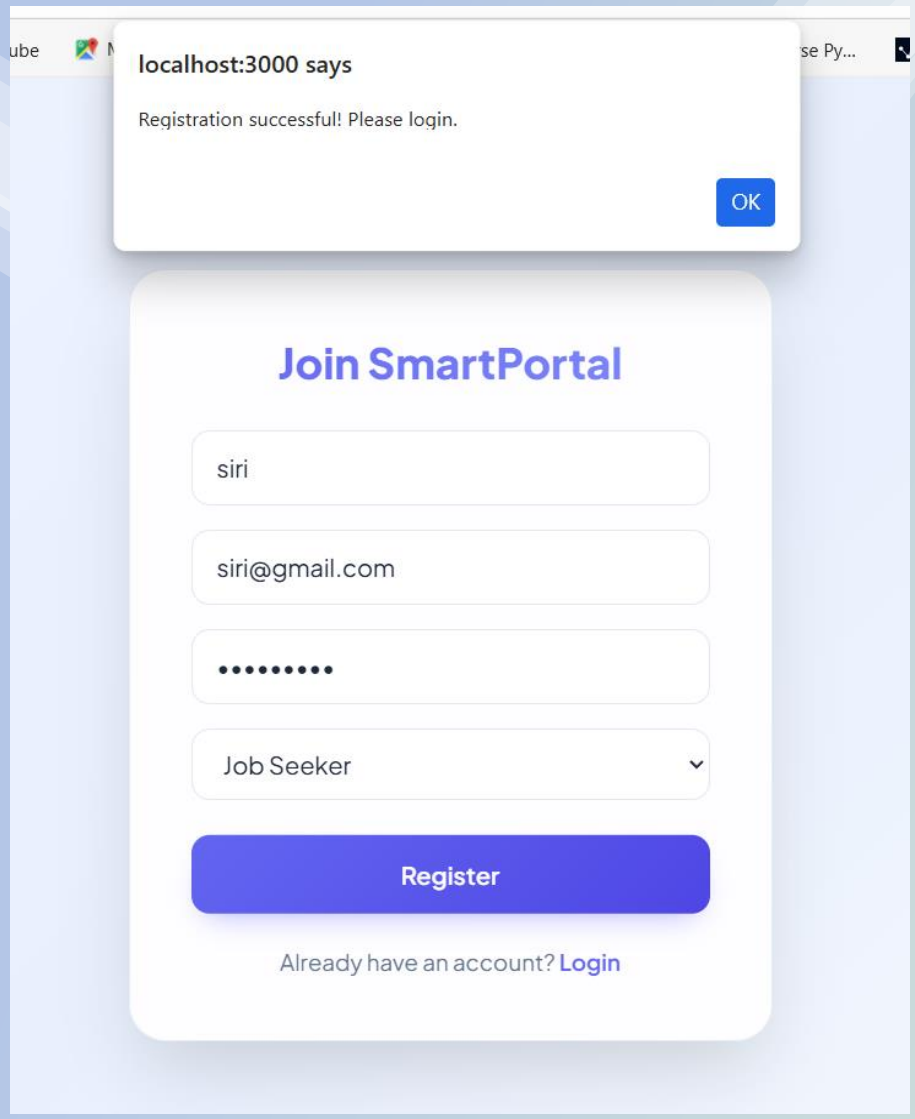
RESULTS - JOB SEEKER SIDE MODEL

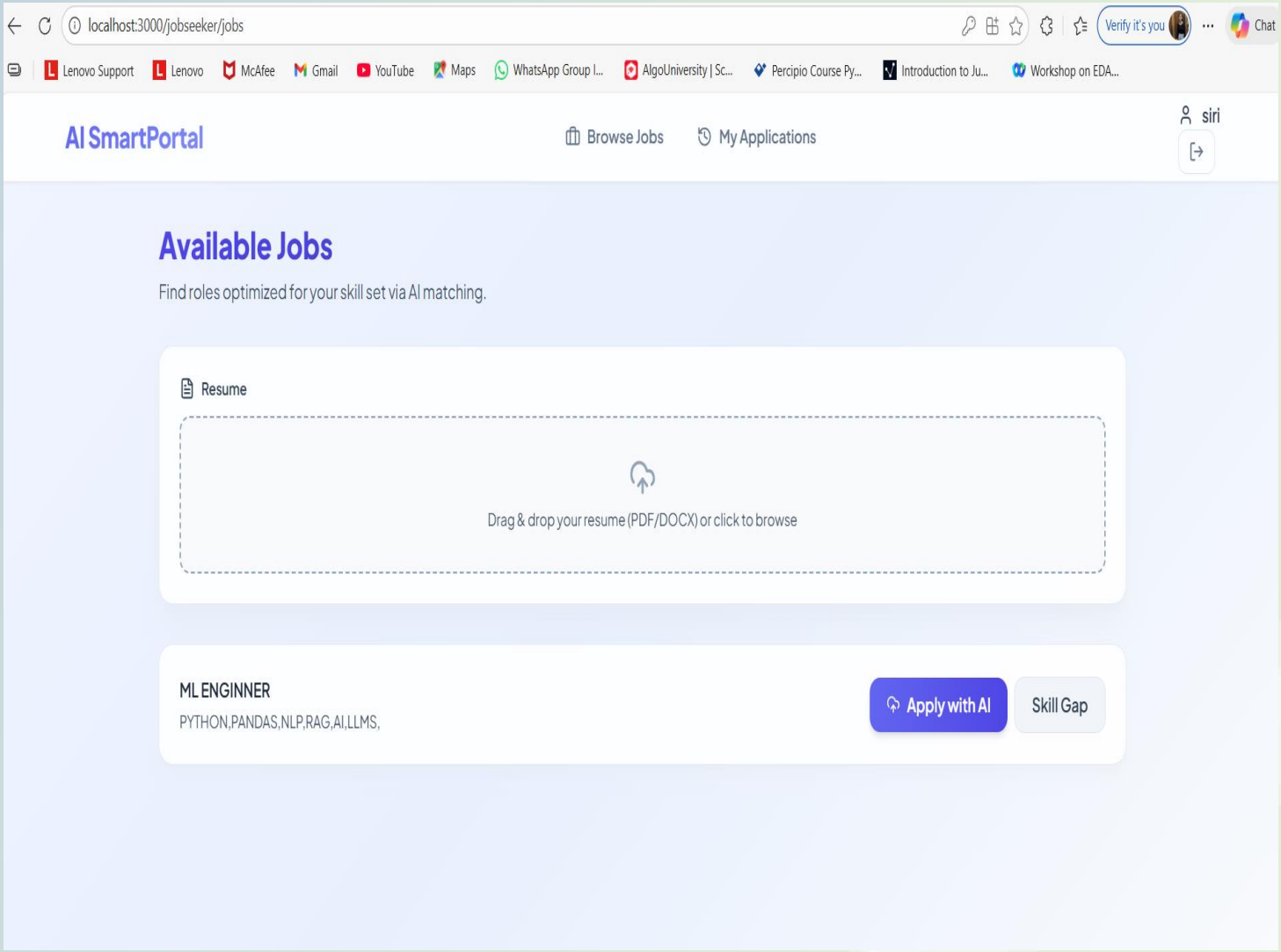
--- Model Accuracy ---
Logistic Regression: 66.55%
Random Forest: 89.57%
SBERT Accuracy: 85.95%

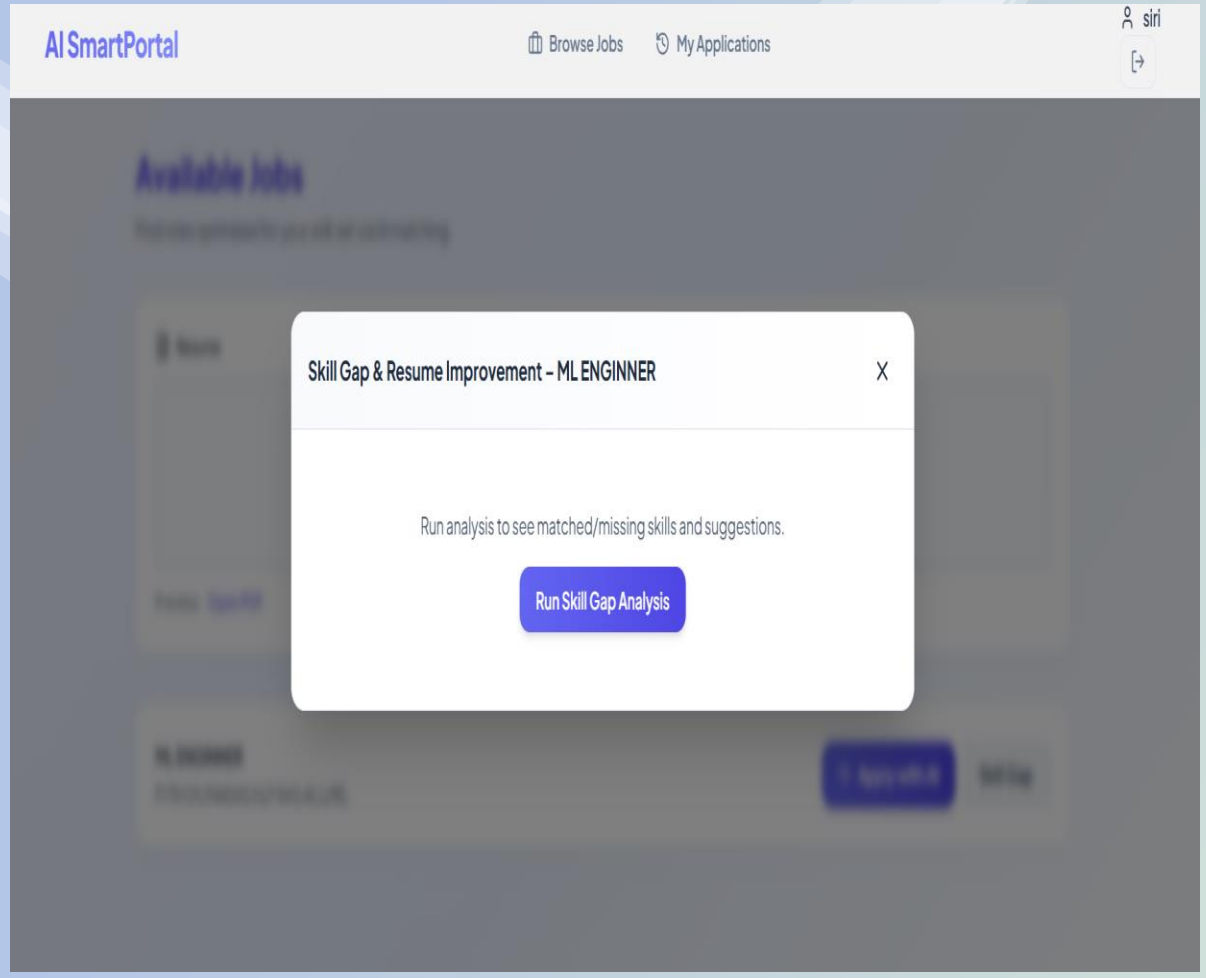
🏆 Best Model Selected: Random Forest
Best Accuracy: 89.57%

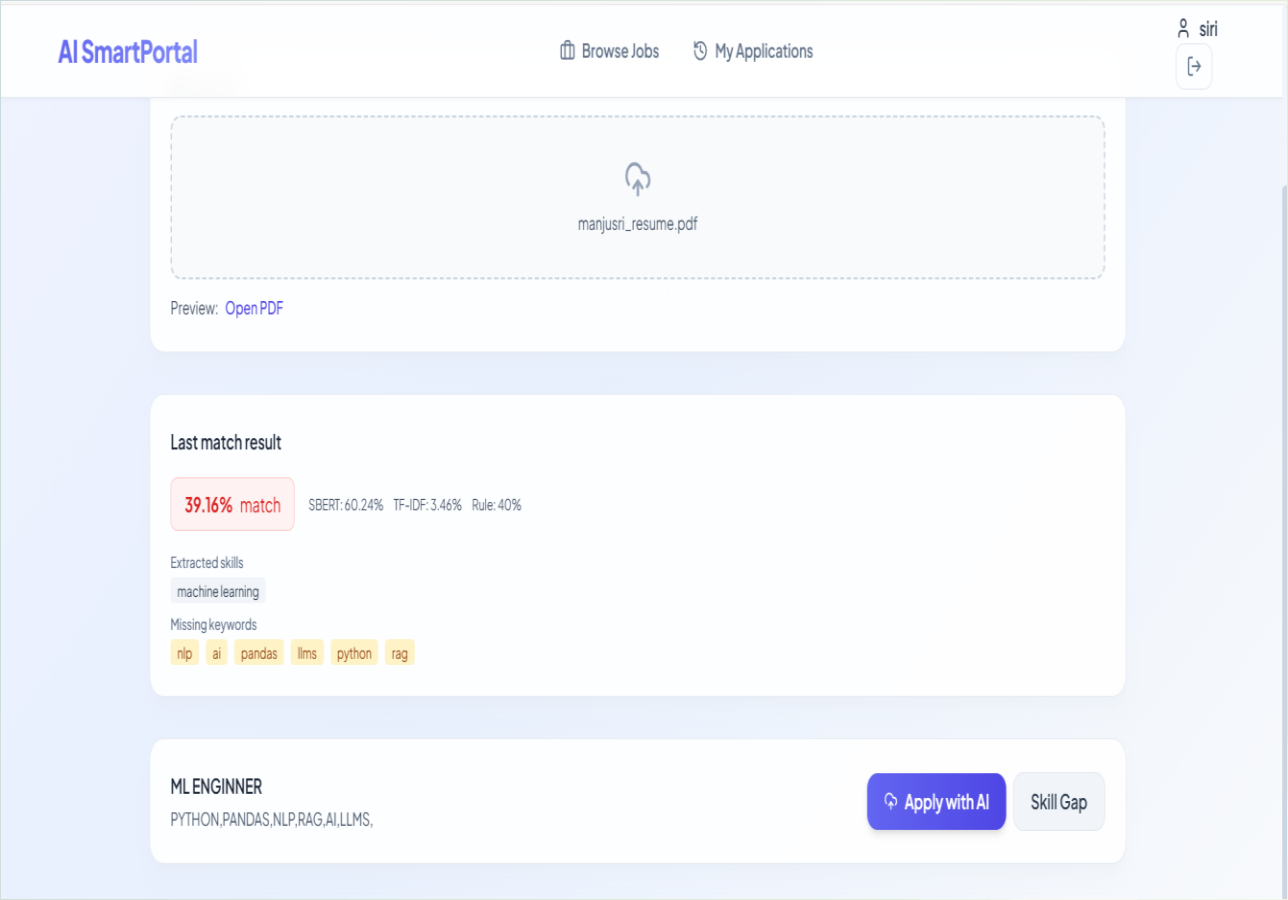
--- Hybrid Matching Model (50–30–20 Weighted Scoring) ---
Final Score = (0.50 × SBERT Semantic Similarity) + (0.30 × TF-IDF Keyword Similarity) + (0.20 × Skill Coverage Score)

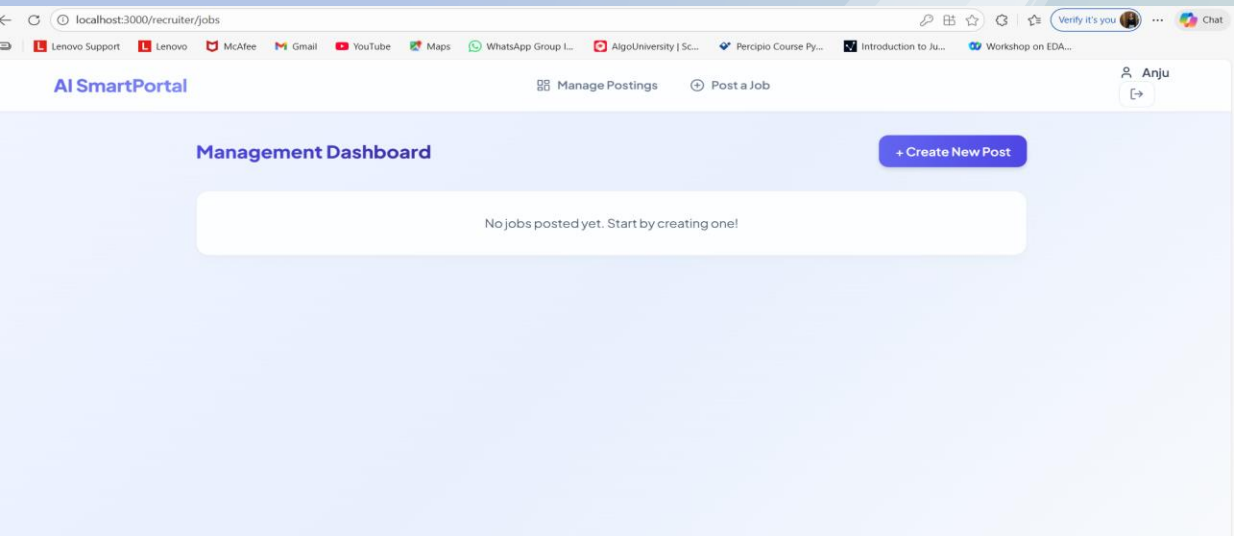
--- Analyzing Resume...
Detected Experience Level: **Junior**
========= TOP 5 JOB RECOMMENDATIONS =========
#1 Mid-Level Data Analyst
Match Score: **35.53%**
Missing Skills: python; sql; statistics; regression analysis
#2 Mid-Level Data Analyst
Match Score: **35.53%**
Missing Skills: python; sql; statistics; regression analysis
#3 Back-End Intern
Match Score: **35.33%**
Missing Skills: node.js; express; mongodb; postman
#4 Back-End Developer (Fresher)
Match Score: **35.20%**
Missing Skills: node.js; express; mongodb; jwt
#5 Mid-Level Data Analyst
Match Score: **34.87%**
Missing Skills: python; sql; statistics; regression analysis

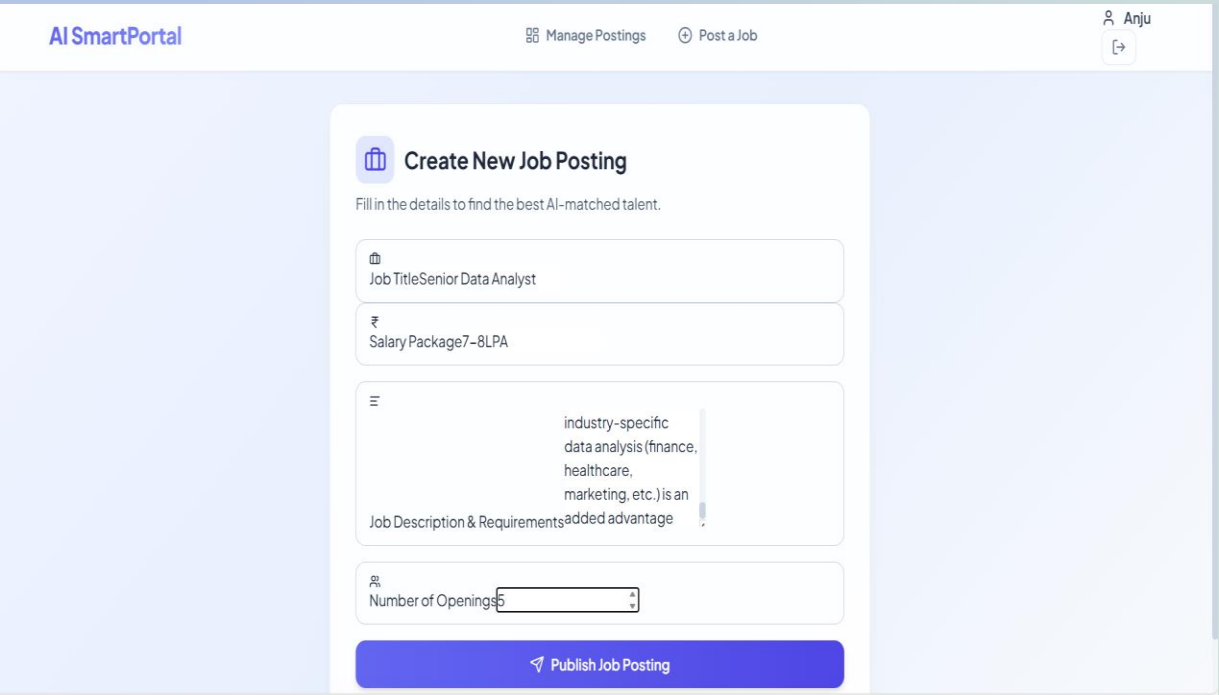

localhost:3000 says
Registration successful! Please login.
OK
Join SmartPortal
siri
siri@gmail.com
Job Seeker
Register
Already have an account? Login

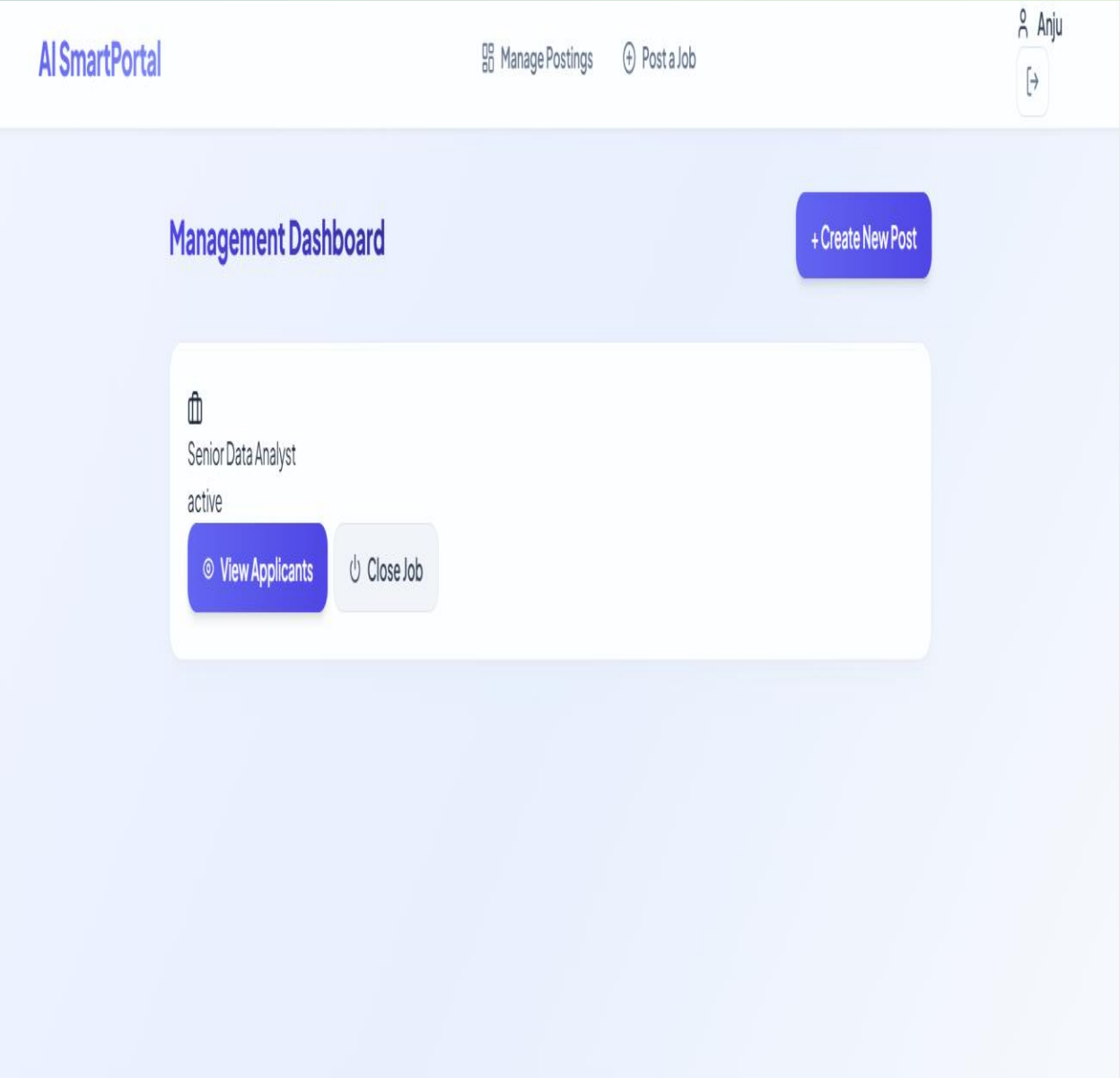

localhost:3000/jobseeker/jobs
AI SmartPortal
Browse Jobs
My Applications
siri
Available Jobs
Find roles optimized for your skill set via AI matching.
Resume
Drag & drop your resume (PDF/DOCX) or click to browse
ML ENGINNER
PYTHON,PANDAS,NLP,RAG,AI,LLMS,
Apply with AI
Skill Gap


AI SmartPortal
Browse Jobs
My Applications
siri
Available Jobs
Skill Gap & Resume Improvement - ML ENGINNER
X
Run analysis to see matched/missing skills and suggestions.
Run Skill Gap Analysis


AI SmartPortal
Browse Jobs
My Applications
siri
manjusri_resume.pdf
Preview: Open PDF
Last match result
39.16% match
SBERT: 60.24% TF-IDF: 3.46% Rule: 40%
Extracted skills
machine learning
Missing keywords
nlp ai pandas llms python rag
ML ENGINNER
PYTHON,PANDAS,NLP,RAG,AI,LLMS,
Apply with AI
Skill Gap


localhost:3000/recruiter/jobs
AI SmartPortal
Manage Postings
Post a Job
Anju
Management Dashboard
+ Create New Post
No jobs posted yet. Start by creating one!


AI SmartPortal
Manage Postings
Post a Job
Anju
Create New Job Posting
Fill in the details to find the best AI-matched talent.
Job TitleSenior Data Analyst
Salary Package7-8LPA
Job Description & Requirements
industry-specific data analysis (finance, healthcare, marketing, etc.) is an added advantage
Number of Openings5
Publish Job Posting


AI SmartPortal
Manage Postings
Post a Job
Anju
Management Dashboard
+ Create New Post
Senior Data Analyst
active
View Applicants
Close Job

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